

Assignment 3

Financial Econometrics: ECO764

Instructor: Wasim Ahmad

Assignment Instructions

This is an *individual assignment*. Students are required to carefully study the case titled “*Parliamentary Election Impact on Indian Capital Markets*”. The case examines possible market outcomes under different electoral scenarios and formulates appropriate option strategies.

The objective of this assignment is to assess your ability to apply derivatives concepts, particularly options pricing, strategy design, and hedging, to a real-world macro-financial event. Your responses should demonstrate analytical clarity, appropriate use of data, and sound financial reasoning.

Submission Guidelines:

- This is an **individual submission**. Any form of plagiarism or collaboration will be strictly penalized.
- Submit your assignment in **PDF format**.
- File name should follow the format: *RollNo_Name_Assignment3.pdf*.
- Ensure that all calculations and explanations are clearly presented.

Submission Details:

- **Last Date:** April 21, 2026 (midnight)
- **Submission Mode:** Email
- **Email ID:** *wasimad24@gmail.com*

Questions

1. Create options strategies for May 15, 2009, based on the political and market scenario prevailing in India at that time. Use the options data for May 15, 2009 provided in Case Exhibit 3.
2. Evaluate the performance of the options strategies constructed in Question 1 using the relevant data from Case Exhibit 3.
3. Examine the hedging effectiveness of stock options using the data on two highly liquid stocks provided below.

DATA FOR OPTION STOCK ICICI BANK

Call American — May 15, 2009

Strike	Open	High	Low	Close	LTP	Contracts	Turnover	OI	ΔOI	Underlying
540	36	60	36	58.1	58.5	341	140.041	172900	-34300	574.7
560	29.7	46	27	43.65	45	953	398.008	632100	32900	574.7
580	19.9	34.65	19.5	32.55	31.75	803	341.627	1328600	9100	574.7
600	14	25	14	23.7	23	451	195.839	194600	-7000	574.7
620	9	18.9	9	17.05	18.9	241	107.140	240100	9800	574.7
19-May-09										
540	160	252	150	248	248	113	59.980	123200	-49700	756.15
560	130	234.65	130	197.15	194.05	251	132.835	541800	-90300	756.15
580	128.95	219	64.8	174.3	174.5	275	144.695	1280300	-48300	756.15
600	149	199	77	169.75	163.4	245	127.726	158900	-35700	756.15
620	81	180	70.1	147.3	136.15	190	100.332	211400	-28700	756.15

Put American — May 15, 2009

Strike	Open	High	Low	Close	LTP	Contracts	Turnover	OI	ΔOI	Underlying
500	18	18	10.95	13	13.4	481	173.037	649600	77700	574.7
510	18.5	18.95	14	14.9	14.75	37	13.633	107800	700	574.7
520	22	22	15.25	16.7	17	211	79.469	176400	37100	574.7
540	28.55	29	19.95	22.2	23	459	181.332	438200	120400	574.7
560	38	38	26	28.7	29.1	526	217.837	280000	207200	574.7
19-May-09										
500	2	3	1	1.6	1.85	276	96.905	667800	18200	756.15
510	1.2	1.9	1	1.6	1.65	41	14.680	107100	-700	756.15
520	3.5	4	1	1.65	2.45	95	34.724	155400	-21700	756.15
540	4	5	0.7	2.15	2.45	126	47.833	396900	-41300	756.15
560	5	5.1	1.7	2.7	2.5	183	72.138	253400	-27300	756.15

DATA FOR OPTION STOCK RELIANCE

Call American — May 15, 2009

Strike	Open	High	Low	Close	LTP	Contracts	Turnover	OI	ΔOI	Underlying
1890	113.95	130	100.5	122.9	130	86	51.637	91500	3600	1950.7
1920	94	115	82.05	107.05	114	661	399.531	153600	18300	1950.7
1950	79.6	96	68	89.75	95	547	332.995	344700	0	1950.7
1980	62	80	54.15	72.15	80	408	250.21	376200	-9600	1950.7
2010	50.9	66	43.05	60.85	66	268	165.961	60000	1500	1950.7
19-May-09										
1890	550	550	360	365	370	91	63.169	72600	-18900	2230.9

1920	330	440	311.25	333.35	350	187	129.734	121200	-32400	2230.9
1950	400	410	210	306	300	105	72.553	338100	-6600	2230.9
1980	299	417	200	275.15	265.5	229	157.921	361200	-14400	2230.9
2010	241	350	241	243.75	242	101	69.515	48000	-12000	2230.9

Put American — May 15, 2009

Strike	Open	High	Low	Close	LTP	Contracts	Turnover	OI	ΔOI	Underlying
1830	47.55	50	40	43	40.5	63	35.446	38400	600	1950.7
1860	60	62	47.2	51.1	48	158	90.75	83400	3000	1950.7
1890	72	73	56.1	61.5	56.1	260	152.459	135300	24600	1950.7
1920	87.9	87.9	68.15	74.1	71	570	341.479	127800	62400	1950.7
1950	91.7	96	80.1	85.35	84	141	86.13	59700	20700	1950.7
19-May-09										
1830	0.1	6.8	0.1	6.2	5	24	13.208	36900	-1500	2230.9
1860	8	9	3.5	6.95	6.15	89	49.816	76500	-6600	2230.9
1890	15	15	4	8.25	8.95	179	101.833	111600	-23700	2230.9
1920	20	20	4.1	11.4	12.25	283	163.693	115200	-12600	2230.9
1950	25	25	2.2	13.75	15.75	125	73.476	69300	9600	2230.9

4. Explore the return potential of key stocks of the Indian capital markets. Case Exhibit 5 can be utilized for this, as close analysis of case Exhibit 5 clearly shows that not all stocks perform in tandem with the markets. You may also give the reasons for this.
5. Formulate the options strategies in a scenario where Nifty either hits the lower circuits, plunges heavily or remains highly volatile, like the one presented in case Exhibit 2.

Source: NSE, India.